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SUPPLEMENTARY INFORMATION

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Committees with implicit biases promote fewer women when they do not believe gender bias exists

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Supplementary Results

Descriptive statistics and gender differences on individual level data

The magnitude of the implicit Gender-Science bias (IAT score) computed across participants (M = 0.38, SD = 0.44, $CI_{95\%}$ [0.33, 0.42]) was similar to the average IAT score in a larger French sample[18] (M = 0.42, SD = 0.43, N = 5810, $CI_{95\%}$ [0.41, 0.44]). Supplementary Table 3 provides means and standard deviations of individual level data by gender. Notably, men exhibited a stronger implicit Science = Male bias (M = 0.44, SD = 0.40) than did women, (M = 0.28, SD = 0.48), $t(281.21) = 3.42$, $P < 0.001$, Cohen's d effect size (d) = 0.36 (independent sample Student's t-test, two-tailed). However, a comparison of both means against 0 (one-sample t-tests) indicated that both men, $t(253) = 17.29$, $P < 0.001$, and women, $t(153) = 7.24$, $P < 0.001$, found it easier to associate science with male (and arts with female) compared to the reverse.

When rating the individual factors that may contribute to gender disparities in STEM fields, participants attributed women's underrepresentation more to family constraints (M = 4.31, SD = 1.20), discrimination (M = 3.58, SD = 1.14), and personal choice (M = 3.18, SD = 1.04), than to gender differences in ability (M = 1.52, SD = 0.71). Men gave significantly lower attributions than did women to discrimination, t (406) = -5.66, $P < 0.001$, and personal choice, t (406) = -2.55, P = 0.011, marginally higher attributions to gender differences in ability, t (406) = 1.73, P = 0.085, and marginally lower attributions to family constraints, t (404) = -1.90, P = 0.058. Additionally, concerning participants' expectancies for men's and women's success in their scientific field, participants reported higher probabilities of success for men than for women. Independent sample t-tests (two-tailed, Ps < 0.05) indicated no significant difference in men's and women's expectations for men's success but showed that male participants provided lower probabilities for women's success than did female participants (Ps < 0.05).

Descriptive statistics on committee level data

Supplementary Table 1 provides means and standard deviations of committee-level variables. As noted in the main text, committees exhibited implicit gender biases as measured by the IAT, suggesting that committee members exhibited a general tendency to cognitively associate science more with male than with female. There was substantial variation in committees' IAT scores (range: -0.07 to 0.67). This variation was not significantly related to the percentage of women on the committee (r = -0.10, P = 0.554). There was also no significant difference in committee-level IAT scores between the physical/math committees (#section = 1- 20, M = 0.35, SD = 0.16) and life/social sciences committees (#section = 21-40, M = 0.36), SD = 0.14), t (37) = -0.27, P = 0.788 (see note to Supplementary Table 1 for committee numbers).

Descriptive patterns of selection decisions at Year 1 and Year 2

Supplemental analyses tested the interactive effect of implicit associations and explicit beliefs on selection decisions made at Year 1 and Year 2 in two separate analyses. Although no effects were significant, the descriptive patterns of data plotted in Supplementary Figure 1 show somewhat different (non-significant) relationships with selection decisions in each year. At Year 1 (panel a), there was a small non-significant pattern suggesting a slight tendency for committees with stronger implicit biases paired with a lower belief that barriers are a problem to be most likely to favor women in their selection decisions in Year 1. This was not an anticipated pattern, but one that suggests that those committees with greater biases might have been somewhat cautious in their decision-making because of the salience of accountability efforts by the organization. However, these committees also showed the largest change in their selection outcomes in the direction of disadvantaging women at Year 2 (see panels b and c). In contrast, committees who endorsed external barriers as a problem made decisions closer

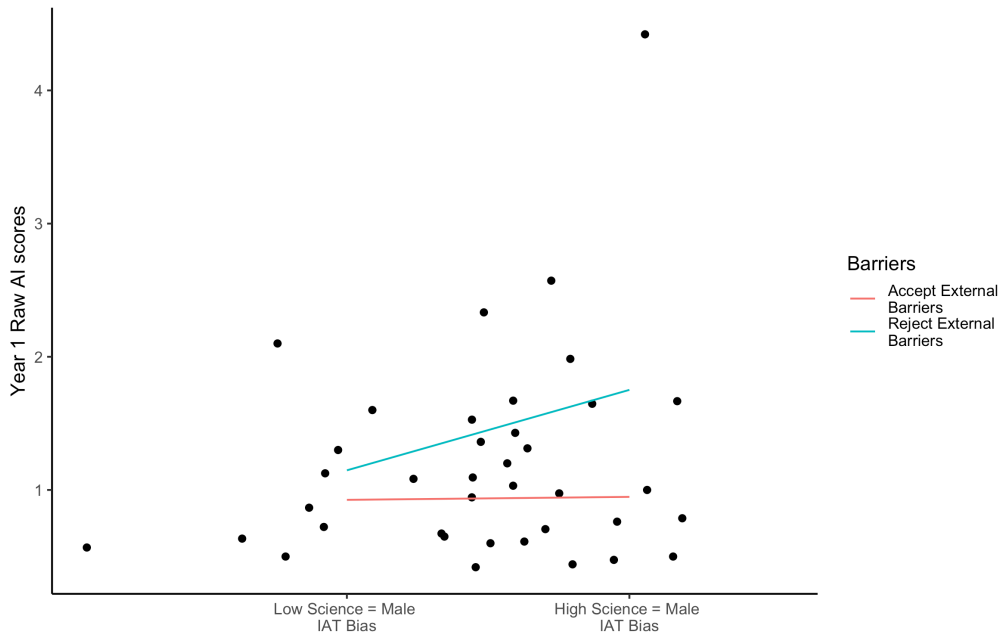
to parity at both years (see panels a, b, and c), whatever their implicit bias, in line with hypotheses.

Outliers in focal analyses

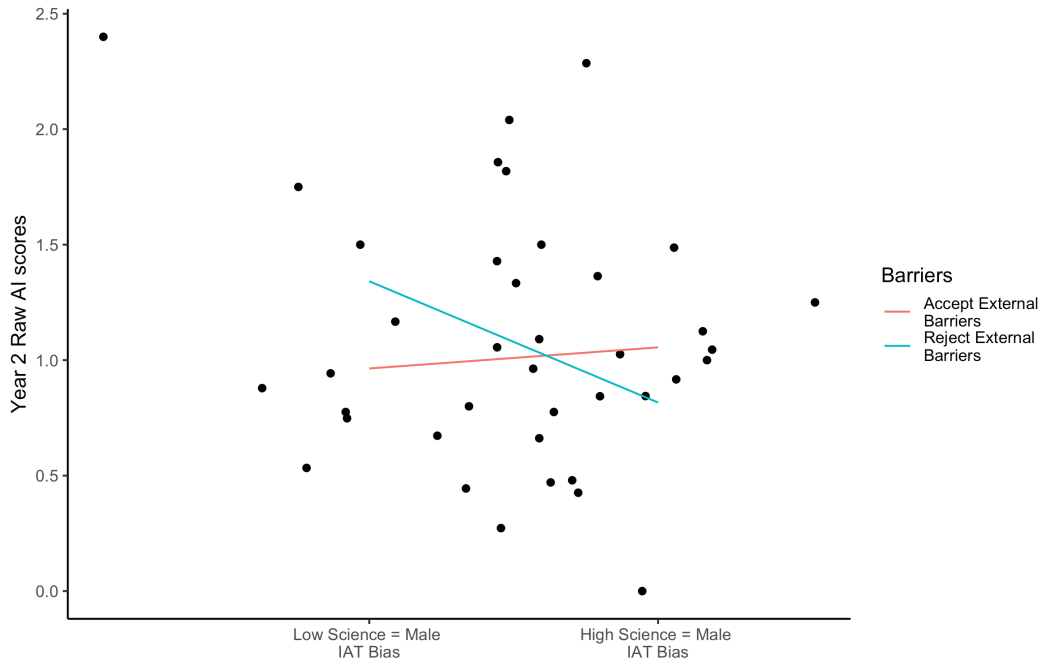
Two committees were identified as being possible multivariate outliers (#13, Physical Chemistry; #1, Math). We retained these outliers in focal analyses because their values (especially on selection outcomes) were not due to measurement error but instead reflected true extreme values in the sample. However, we conducted additional 95% percentile bootstrap confidence intervals of the interactions and the key simple slope estimating the predictive effects of implicit bias on selection decisions at low levels of the moderator where biases were found to predict decisions. Supplementary Table 4 summarizes these results with the composite measure of beliefs as a moderator and on each of the ways of operationalizing selection decisions (raw and log transformed AI ratio, d score pass-rates, and gender asymmetry). The effect sizes were always in the predicted direction for both the interaction and the simple slope for IAT predicting selections when committees discounted external barriers for women's underrepresentation.

Supplementary Figures

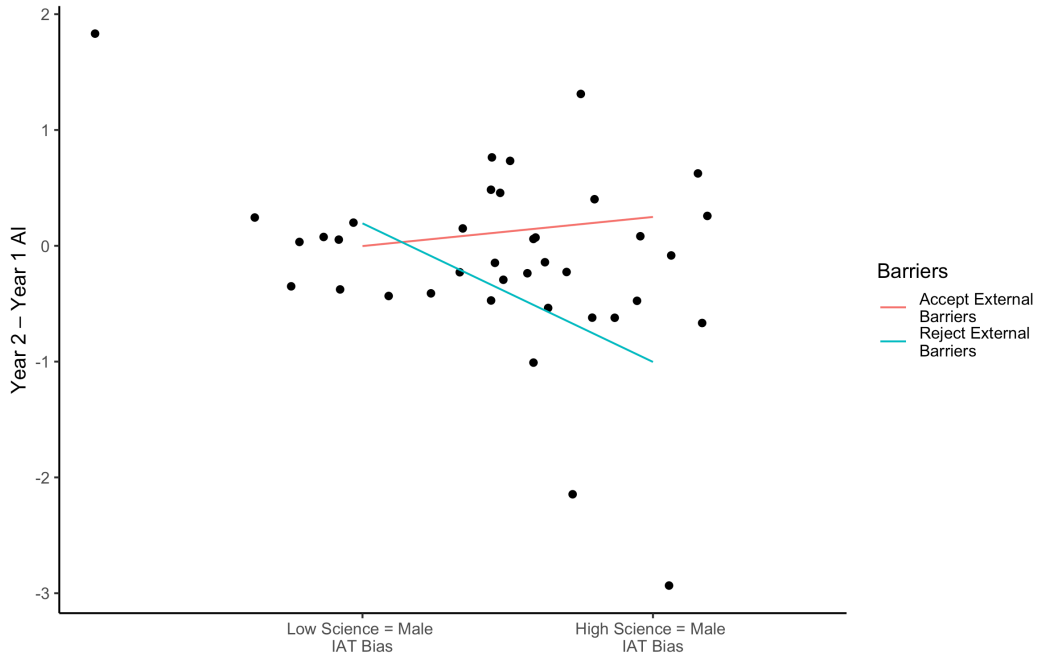
a



b



c



Supplementary Figure 1. Descriptive data for the gender IAT by explicit beliefs interaction on raw AI scores at Year 1 (panel a), Year 2 (panel b), and Year 2 – Year 1 (panel c) .

Supplementary Tables

Excel files:

Supplementary Table 1. Means (standard deviations) of committee-level IAT score and self-reported variables

Supplementary Table 2. Analyses on log transformed AI Ratios Examining other Covariates

	AI Ratios (N = 37) Without additional covariate				**AI Ratios (N = 37) Cov= % women on committee**				**AI Ratios (N = 37) Cov= type of science (0 = physical; 1 = life/social)**			
	β	*t*	***P***	CI_{95}	***β***	***t***	***P***	CI_{95}	***β***	***t***	***P***	CI_{95}
Constant	-0.06	-0.40	0.695	-0.37, 0.25	-0.05	-0.35	0.732	-0.37, 0.26	-0.06	-0.40	0.694	-0.37, 0.25
Year 1 baseline	0.44	2.63	0.013	0.10, 0.78	0.43	2.48	0.019	0.08, 0.78	0.44	2.61	0.014	0.10, 0.78
Covariate	na	na	na	na	-0.06	-0.30	0.767	-0.44, 0.32	0.16	0.90	0.377	-0.20, 0.52
IAT Bias	-0.14	-0.81	0.422	-0.48, 0.21	-0.13	-0.74	0.462	-0.48, 0.22	-0.13	-0.77	0.448	-0.47, 0.21
External barriers	0.05	0.26	0.796	-0.33, 0.42	0.07	0.36	0.721	-0.35, 0.50	-0.03	-0.16	0.871	-0.46, 0.39
Interaction	**0.38**	**2.30**	**0.028**	**0.04, 0.72**	**0.38**	**2.26**	**0.031**	**0.04, 0.73**	**0.39**	**2.34**	**0.026**	**0.05, 0.73**
R^2	0.28		0.029		0.28		0.058		0.30		0.043	
Simple slope for IAT bias												
Low belief in external barriers	**-0.61**	**-2.42**	**0.022**	**-1.13,-0.10**	**-0.60**	**-2.33**	**0.027**	**-1.13, -0.07**	**-0.62**	**-2.43**	**0.021**	**-1.14, -0.10**
High belief in external barriers	0.13	0.60	0.555	-0.31, 0.56	0.13	0.62	0.541	-0.31, 0.58	0.14	0.66	0.516	-0.30, 0.58

Supplementary Table 3. Descriptive statistics on individual level data from male and female respondents

Variables	Men		Women		*t*	*P*
	Mean	SD	Mean	SD		
1. IAT	0.44	0.40	0.28	0.48	3.57	0.001
2. Discrimination	3.35	1.11	3.99	1.07	-5.67	0.001
3. Personal choice	3.09	1.00	3.36	1.09	-2.55	0.011
4. Gender differences in ability	1.56	0.71	1.44	0.69	1.73	0.085
5. Family constraints	4.21	1.17	4.45	1.25	-1.90	0.058
6. Men's success	4.79	0.97	4.86	0.98	-0.78	0.438
7. Women's success	4.54	0.99	4.82	0.96	-2.74	0.006

Supplementary Table 4. Table of percentile bootstrap confidence intervals (with 10,000 resamples) of focal interactions and key simple slope to assess robustness of effects to outliers.

	Operationalization of Selection Decisions											
	Log Transformed Adverse Impact Ratios (N = 37)			**Raw Adverse Impact Ratios** (N = 38)			**d_score Pass-Through Rates** (N = 37)			**Asymmetry Scores (Men selected – Women selected)** (N = 39)		
	β	Boot Mean (SE)	CI_{95}	β	Boot Mean (SE)	CI_{95}	β	Boot Mean (SE)	CI_{95}	β	Boot Mean (SE)	CI_{95}
IAT x External Barriers	**0.38**	**0.36 (0.14)**	**0.07, 0.63**	**0.49**	**0.47 (0.15)**	**0.16, 0.75**	**0.38**	**0.37 (0.16)**	**0.05, .67**	**-0.22**	**-0.24 (.12)**	**-0.55, -0.06**
IAT at –1 SD	-0.52	-0.48[1] (0.22)	-0.84, 0.02	**-0.73**	**-0.67 (0.21)**	**-0.94,-0.27**	-0.50	-0.45[1] (0.24)	-0.86, 0.07	**0.36**	**0.36 (0.12)**	**0.06, 0.56**
IAT at +1 SD	0.24	0.25 (0.17)	-0.06, 0.63	0.25	0.26 (0.18)	-0.04,0.71	0.26	0.28 (0.19)	-0.05, 0.73	-0.07	-0.13 (0.20)	-0.64, 0.15

Note. Asymmetry is coded so that higher numbers mean favoring men over women, whereas adverse impact and d_score pass through rates are interpreted as lower numbers imply favoring men over women. [1] These coefficients fall within a 90% confidence interval.